Design report

Solution Configurator, residential system

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Site and inputs

Postcode	80538 München
Position	48.140, 11.588
Annual consumption	5,000 kWh
PV array	10 kWp
Roof plane 1	10 kWp, tilt 35°, facing 180°
Grid connection	3-phase
Backup supply	Not needed
Heat pump	No
Electric car	No

PV yield

Annual yield, range over 2005–2023	10,034 – 12,164 kWh
Mean annual yield	11,190 kWh
Specific yield	1,003 – 1,216 kWh/kWp
Source	PVGIS 5 (PVGIS-SARAH3) through pvlib, system loss 14 %

Storage sizing

PV at least 0.5 kWp per MWh of annual consumption (HTW Berlin)	2 kWp per MWh, at least 0.5 · met
Usable capacity at most 1.5 kWh per kWp (HTW Berlin)	1.5 kWh × 10 kWp = 15 kWh
Usable capacity at most 1.5 kWh per MWh of annual consumption (HTW Berlin)	1.5 kWh × 5 MWh = 7.5 kWh
Usable capacity, HTW upper limit	7.5 kWh
Proposed usable capacity	5.8 kWh (1 × GW6.0-BAT-D-G20)

Self-sufficiency and self-consumption

Self-sufficiency with storage	61 % – 71 %
Self-consumption with storage	33 % – 38 %
Self-sufficiency without storage	31 % – 41 %
Self-consumption without storage	15 % – 20 %
Method	HTW Berlin Unabhängigkeitsrechner tables; the range covers the spread of real households (most within 5 percentage points of the table value)

Yield: PVGIS © European Union · Self-sufficiency: HTW Berlin · Postcodes: GeoNames (CC BY 4.0)

Selection basis

Inverter	GoodWe GW10K-ETA-G20
Application	residential, as selected · met
Grid connection phases	3-phase, as selected · met
Maximum PV input of the inverter covers the array	20 kW for a 10 kWp array · met
DC/AC ratio within the manufacturer's limit	1.00, at most 2.00 per the manufacturer · met
Array fits the MPPT inputs	22 modules connected · met
Battery released by the manufacturers	6 released batteries · met
String layout	22 × 460 Wp in 2 strings; Voc at −10 °C: 58.3 V per module

Manufacturer releases and documents

GW10K-ETA-G20 + GW6.0-BAT-D-G20	Released
Document	GW_ESA-5-30kW_Datasheet-EN.pdf https://de-linker.com/wp-content/uploads/2026/09/GW_ESA-5-30kW_Datasheet-EN.pdf
Document	GW_ESA-5-30kW_User-Manual-EN_V1.5.pdf https://de-linker.com/wp-content/uploads/2026/10/GW_ESA-5-30kW_User-Manual-EN_V1.5.pdf

The GoodWe ESA datasheet lists GW6.0-BAT-D-G20 and GW9.0-BAT-D-G20 modules and prints scalability of 12 pcs (p. 3). The user manual limits one column to six batteries and names a battery expansion kit for higher capacity or a lower column (p. 79).

Bill of materials

Qty	Item	Model	Supplied by
1 pc	Hybrid inverter	GoodWe GW10K-ETA-G20	Linker product
1 pc	Battery modules	GoodWe GW6.0-BAT-D-G20	Linker product
22 pcs	PV modules of your choice (layout checked with Trina Solar TSM-460NEG9R.28)	–	Installer's scope
1 set	Mounting system for 22 modules	–	Installer's scope
60 m	PV string cable for both poles; measure the route on site	–	Installer's scope
4 pairs	PV connector pairs, mating the module and the inverter connectors	–	Installer's scope
1 set	AC protection and grid connection to VDE-AR-N 4105	–	Installer's scope

Assumptions and limits

This report contains no economics: no payback period, cash flow or feed-in tariff.

The string layout uses Trina Solar TSM-460NEG9R.28 as a reference module; the module you install can change it.

Self-sufficiency figures describe an average household; measured values differ with each household's habits.

Without a roof plane, the yield assumes 35° tilt facing south (HTW Berlin calculator default).

Open-circuit voltage checked at −10 °C cell temperature. (Planning assumption without a published source)

MPP voltage checked at 70 °C cell temperature, with the module's Voc temperature coefficient. (Planning assumption without a published source)

DC cable: 15 m per string and pole until the route is measured. (Planning assumption without a published source)

Yield: PVGIS © European Union · Self-sufficiency: HTW Berlin · Postcodes: GeoNames (CC BY 4.0)

Two connector pairs per string: one at the array, one at the inverter. (Planning assumption without a published source)

Next steps

Add the Linker products to your solution draft and request a Proforma; we reply within one business day.

Before ordering, confirm the string layout with the module you install, the cable routes and the grid connection on site.

Yield: PVGIS © European Union · Self-sufficiency: HTW Berlin · Postcodes: GeoNames (CC BY 4.0)